

## 20 Comparing Two Means

- Most interesting statistical problems involve multiple measured variables. For example, many problems involve comparing two (or more) populations or groups based on two (or more) samples.
- In such situations, each population or group will have its own parameters, and there will often be dependence between parameters. We are often interested in difference or ratios of parameters between groups.

**Example 20.1.** Do newborns born to mothers who smoke tend to weigh less at birth than newborns from mothers who don't smoke? We'll investigate this question using birthweight (pounds) data on a sample of full term births in North Carolina over a one year period.

Assume birthweights follow a conditional Normal distribution with mean  $\mu_1$  for nonsmokers and mean  $\mu_2$  for smokers, and standard deviation  $\sigma$ .

Note: our primary goal will be to compare the means  $\mu_1$  and  $\mu_2$ . We're assuming a common standard deviation  $\sigma$  to simplify a little, but we could (and probably should) also let standard deviation vary by smoking status. (We could also assume something other than conditional Normal, but again, we're simplifying.)

1. Suggest a prior distribution.
2. The prior distribution will be a joint distribution on  $(\mu_1, \mu_2, \sigma)$  triples. We could assume a prior under which  $\mu_1$  and  $\mu_2$  are independent. But why might we want to assume some prior *dependence* between  $\mu_1$  and  $\mu_2$ ? (For some motivation it might help to consider what the frequentist null hypothesis would be.)

3. Regardless of your answers to the previous parts, assume for the prior:  $\mu_1, \mu_2, \sigma$  are independent; each of  $\mu_1, \mu_2$  has a Normal(7.5, 0.5) distribution;  $\sigma$  has a Half-Normal(0, 1) distribution. How do you interpret the parameter  $\mu_1 - \mu_2$ ? Plot the prior distribution of  $\mu_1 - \mu_2$ , and find prior central 50%, 80%, and 98% credible intervals. Also compute the prior probability that  $\mu_1 - \mu_2 > 0$ .
4. Briefly describe the sample data, summarized in Section [20.1.2](#).
5. Is it reasonable to assume that the two *samples* are independent? (In this case the  $y_1$  and  $y_2$  samples would be *conditionally independent* given  $(\mu_1, \mu_2, \sigma)$ .)
6. Describe how you would compute the likelihood. For concreteness, how you would you compute the likelihood if there were only 4 babies in the sample: 2 non-smokers with birthweights of 8 pounds and 7 pounds, and 2 smokers with birthweights of 8.3 pounds and 7.1 pounds.
7. Use `brms` to approximate the posterior distribution. Are  $\mu_1$  and  $\mu_2$  independent according the posterior? Why do you think that is?

8. Plot the posterior distribution of  $\mu_1 - \mu_2$  and describe it. Compute and interpret posterior central 50%, 80%, and 98% credible intervals. Also compute and interpret the posterior probability that  $\mu_1 - \mu_2 > 0$ .
9. If we're interested in  $\mu_1 - \mu_2$ , why didn't we put a prior directly on  $\mu_1 - \mu_2$  rather than on  $(\mu_1, \mu_2)$ ?
10. Plot the posterior distribution of  $\mu_1/\mu_2$ , describe it, and find and interpret posterior central 50%, 80%, and 98% credible intervals.
11. Is there some evidence that babies whose mothers smoke tend to weigh less than those whose mothers don't smoke?
12. Can we say that smoking is the cause of the difference in mean weights?
13. Is there some evidence that babies whose mothers smoke tend to weigh *much* less than those whose mothers don't smoke? Explain.

14. One quantity of interest is the **effect size**, which is a way of measuring the magnitude of the difference between groups. When comparing two means, a simple measure of effect size (*Cohen's d*) is

$$\frac{\mu_1 - \mu_2}{\sigma}$$

Plot the posterior distribution of this effect size and describe it. Compute and interpret posterior central 50%, 80%, and 98% credible intervals.

### Independence in the data versus in the prior/posterior

It is typical to assume *independence in the data*, e.g., independence of values of the measured variables within and between samples (conditional on the parameters). Whether independence in the data is a reasonable assumption depends on how the data is collected.

But whether it is reasonable to assume prior *independence of parameters* is a completely separate question and is dependent upon our subjective beliefs about any relationships between parameters. In hierarchical models, independence between parameters is often implied by the hierarchical structure.

### Transformations of parameters

The primary output of a Bayesian data analysis is the full joint posterior distribution on all parameters. Given the joint distribution, the distribution of transformations of the primary parameters is readily obtained.

### Effect size for comparing means

When comparing groups, a more important question than “is there a difference?” is “*how large* is the difference?” An **effect size** is a measure of the magnitude of a difference between groups. A difference in parameters can be used to measure the absolute size of the difference in the measurement units of the variable, but effect size can also be measured as a relative difference.

When comparing a numerical variable between two groups, one measure of the population effect size is **Cohens's d**

$$\frac{\mu_1 - \mu_2}{\sigma}$$

The values of any numerical variable vary naturally from unit to unit. The SD of the numerical variable measures the degree to which individual values of the variable vary naturally, so the SD

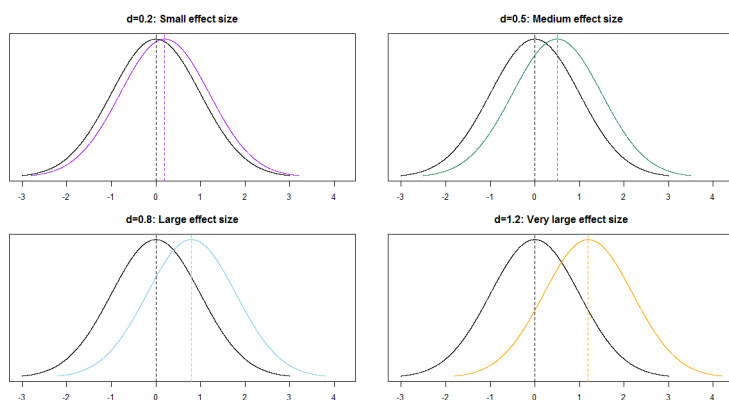
provides a natural “scale” for the variable. Cohen’s  $d$  compares the magnitude of the difference in means relative to the natural scale (SD) for the variable.

Some rough guidelines for interpreting  $|d|$ :

| $d$         | 0.2   | 0.5    | 0.8   | 1.2        | 2.0  |
|-------------|-------|--------|-------|------------|------|
| Effect size | Small | Medium | Large | Very Large | Huge |

For example, assume the two population distributions are Normal and the two population standard deviations are equal. Then when the effect size is 1.0 the median of the distribution with the higher mean is the 84th percentile of the distribution with the lower mean, which is a very large difference.

| $d$                                                             | 0.2   | 0.5    | 0.8   | 1.0  | 1.2        | 2.0  |
|-----------------------------------------------------------------|-------|--------|-------|------|------------|------|
| Effect size                                                     | Small | Medium | Large |      | Very Large | Huge |
| Median of population 1 is<br>(blank) percentile of population 2 | 58th  | 69th   | 79th  | 84th | 89th       | 98th |



## 20.1 Notes

### 20.1.1 Prior distribution of $\mu_1 - \mu_2$

The prior distribution of  $\mu_1 - \mu_2$  is Normal with mean  $7.5 - 7.5 = 0$  and standard deviation  $\sqrt{0.5^2 + 0.5^2} = 0.707$

```
qnorm(
  c(0.01, 0.10, 0.25, 0.75, 0.90, 0.99),
  mean = 7.5 - 7.5,
  sd = sqrt(0.5^2 + 0.5^2)
)
```

```
[1] -1.6449764 -0.9061938 -0.4769363  0.4769363  0.9061938  1.6449764
```

## 20.1.2 Sample data

```
data = read_csv("baby_smoke.csv")
```

```
Rows: 999 Columns: 13
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (7): mature, premie, marital, lowbirthweight, gender, habit, whitemom
```

```
dbl (6): fage, mage, weeks, visits, gained, weight
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(data)
```

```
# A tibble: 6 x 13
```

|   | fage  | mage  | mature  | weeks | premie | visits | marital | gained | weight  | lowbirthweight |      |         |
|---|-------|-------|---------|-------|--------|--------|---------|--------|---------|----------------|------|---------|
|   | <dbl> | <dbl> | <chr>   | <dbl> | <chr>  | <dbl>  | <chr>   | <dbl>  | <dbl>   | <chr>          |      |         |
| 1 | NA    | 13    | younger | ~     | 39     | full   | ~       | 10     | married | 38             | 7.63 | not low |
| 2 | NA    | 14    | younger | ~     | 42     | full   | ~       | 15     | married | 20             | 7.88 | not low |
| 3 | 19    | 15    | younger | ~     | 37     | full   | ~       | 11     | married | 38             | 6.63 | not low |
| 4 | 21    | 15    | younger | ~     | 41     | full   | ~       | 6      | married | 34             | 8    | not low |
| 5 | NA    | 15    | younger | ~     | 39     | full   | ~       | 9      | married | 27             | 6.38 | not low |
| 6 | NA    | 15    | younger | ~     | 38     | full   | ~       | 19     | married | 22             | 5.38 | low     |

```
# i 3 more variables: gender <chr>, habit <chr>, whitemom <chr>
```

We'll only include full term babies in the analysis. The variables of interest are `weight` and `habit`.

| habit     | n() | mean(weight) | sd(weight) |
|-----------|-----|--------------|------------|
| nonsmoker | 739 | 7.50         | 1.08       |
| smoker    | 107 | 7.17         | 0.97       |

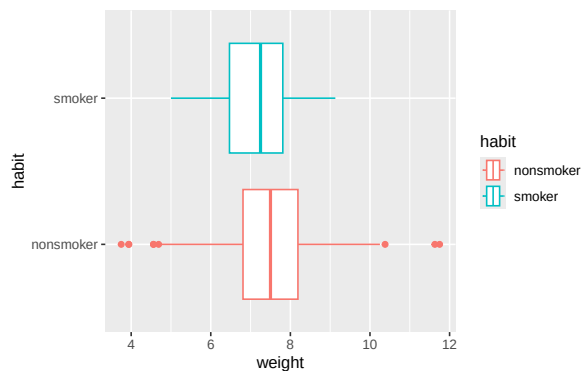
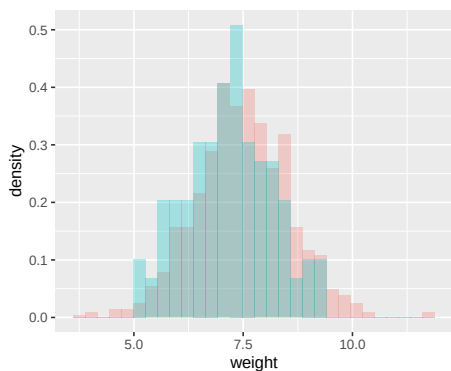
```
data = data |>
  filter(premie == "full term")
```

```
data |>
  group_by(habit) |>
  summarize(n(), mean(weight), sd(weight)) |>
  kbl(digits = 2) |>
  kable_styling()
```

```
ggplot(data, aes(x = weight, fill = habit)) +
  geom_histogram(
    alpha = 0.3,
    aes(y = after_stat(density)),
    position = 'identity'
  )
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.

```
ggplot(data, aes(x = weight, y = habit, col = habit)) +
  geom_boxplot()
```



### 20.1.3 Posterior distribution

```
library(brms)
```

Loading required package: Rcpp

Loading 'brms' package (version 2.23.0). Useful instructions can be found by typing `help('brms')`. A more detailed introduction to the package is available through `vignette('brms_overview')`.

Attaching package: 'brms'

The following objects are masked from 'package:tidybayes':

dstudent\_t, pstudent\_t, qstudent\_t, rstudent\_t

The following object is masked from 'package:stats':

ar

```
library(tidybayes)
library(bayesplot)
```

This is bayesplot version 1.14.0

- Online documentation and vignettes at [mc-stan.org/bayesplot](https://mc-stan.org/bayesplot)
- bayesplot theme set to `bayesplot::theme_default()`
  - \* Does `_not_` affect other ggplot2 plots
  - \* See `?bayesplot_theme_set` for details on theme setting

Attaching package: 'bayesplot'



The following object is masked from 'package:brms':

rhat

We want a posterior for  $\mu_1$  and  $\mu_2$ , so we set the intercept to 0 in the linear model. We can use `get_prior` with our model specifications to see what the parameters will be.

```
get_prior(data = data, family = gaussian(), weight ~ 0 + habit)
```

|                      | prior  | class | coef           | group | resp | dpar | nlpar | lb | ub | tag |
|----------------------|--------|-------|----------------|-------|------|------|-------|----|----|-----|
|                      | (flat) | b     |                |       |      |      |       |    |    |     |
|                      | (flat) | b     | habitnonsmoker |       |      |      |       |    |    |     |
|                      | (flat) | b     | habitsmoker    |       |      |      |       |    |    |     |
| student_t(3, 0, 2.5) | sigma  |       |                |       |      |      |       |    | 0  |     |
| source               |        |       |                |       |      |      |       |    |    |     |
| default              |        |       |                |       |      |      |       |    |    |     |
| (vectorized)         |        |       |                |       |      |      |       |    |    |     |
| (vectorized)         |        |       |                |       |      |      |       |    |    |     |
| default              |        |       |                |       |      |      |       |    |    |     |

Now we fit the model. The `normal(7.5, 0.5)` prior for  $\mu_1$  and  $\mu_2$  will be vectorized.

```
fit <- brm(  
  data = data,  
  family = gaussian(),  
  weight ~ 0 + habit,  
  prior = c(  
    prior(normal(7.5, 0.5), class = b),  
    prior(normal(0, 1), class = sigma)  
  ),  
  iter = 3500,  
  warmup = 1000,  
  chains = 4,  
  refresh = 0  
)
```

Compiling Stan program...

Start sampling

```
prior_summary(fit)
```

```
           prior class           coef group resp dpar nlpar lb ub tag
normal(7.5, 0.5)      b
normal(7.5, 0.5)      b habitnonsmoker
normal(7.5, 0.5)      b   habitsmoker
      normal(0, 1) sigma                      0
      source
      user
(vectorized)
(vectorized)
      user
```

```
summary(fit)
```

```
Family: gaussian
Links: mu = identity
Formula: weight ~ 0 + habit
Data: data (Number of observations: 846)
Draws: 4 chains, each with iter = 3500; warmup = 1000; thin = 1;
      total post-warmup draws = 10000
```

Regression Coefficients:

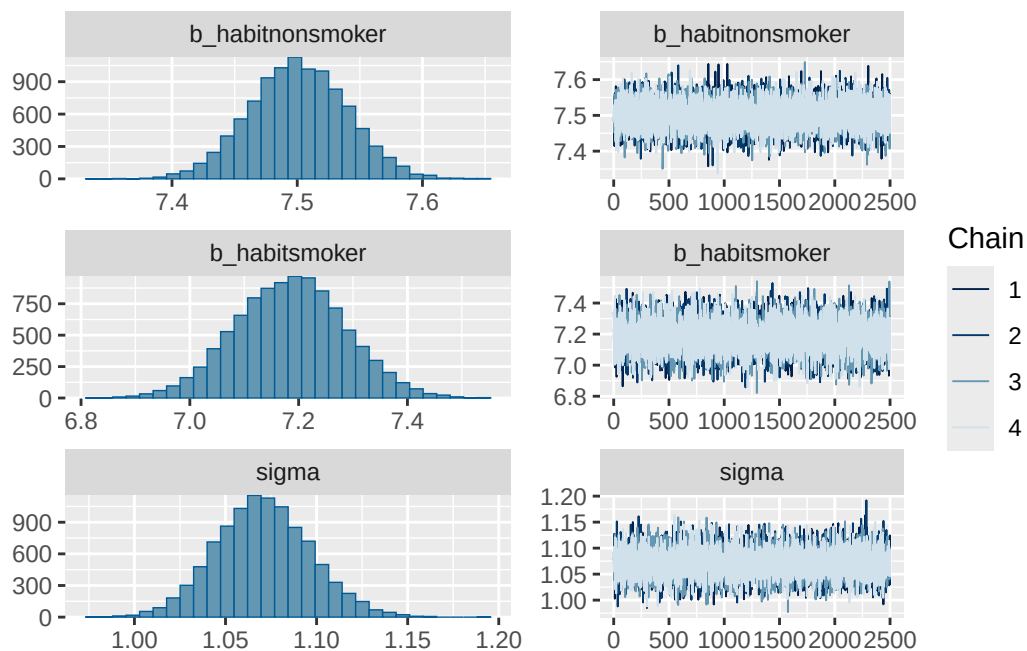
|                | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|----------------|----------|-----------|----------|----------|------|----------|----------|
| habitnonsmoker | 7.50     | 0.04      | 7.43     | 7.58     | 1.00 | 9993     | 7584     |
| habitsmoker    | 7.19     | 0.10      | 6.99     | 7.38     | 1.00 | 8723     | 6750     |

Further Distributional Parameters:

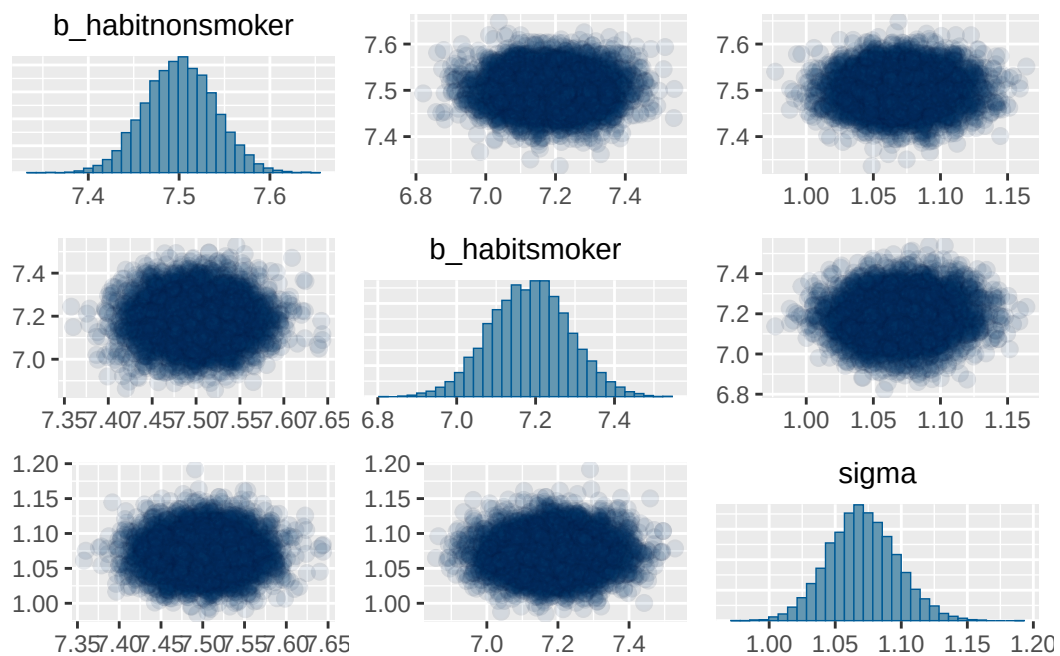
|       | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|-------|----------|-----------|----------|----------|------|----------|----------|
| sigma | 1.07     | 0.03      | 1.02     | 1.12     | 1.00 | 10160    | 7338     |

Draws were sampled using sampling(NUTS). For each parameter, Bulk\_ESS and Tail\_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(fit)
```



```
pairs(fit, off_diag_args = list(alpha = 0.1))
```



| .chain | .iteration | .draw | mu2  | mu1  | sigma |
|--------|------------|-------|------|------|-------|
| 1      | 1          | 1     | 7.34 | 7.49 | 1.09  |
| 1      | 2          | 2     | 7.27 | 7.50 | 1.04  |
| 1      | 3          | 3     | 7.24 | 7.51 | 1.06  |
| 1      | 4          | 4     | 7.23 | 7.50 | 1.05  |
| 1      | 5          | 5     | 7.32 | 7.44 | 1.09  |
| 1      | 6          | 6     | 7.08 | 7.52 | 1.09  |

| .chain | .iteration | .draw | mu2  | mu1  | sigma | mu_diff |
|--------|------------|-------|------|------|-------|---------|
| 1      | 1          | 1     | 7.34 | 7.49 | 1.09  | 0.15    |
| 1      | 2          | 2     | 7.27 | 7.50 | 1.04  | 0.23    |
| 1      | 3          | 3     | 7.24 | 7.51 | 1.06  | 0.27    |
| 1      | 4          | 4     | 7.23 | 7.50 | 1.05  | 0.27    |
| 1      | 5          | 5     | 7.32 | 7.44 | 1.09  | 0.13    |
| 1      | 6          | 6     | 7.08 | 7.52 | 1.09  | 0.45    |

```
posterior = fit |>
  spread_draws(b_habitsmoker, b_habitsnonsmoker, sigma) |>
  rename(mu1 = b_habitsnonsmoker, mu2 = b_habitsmoker)

posterior |> head() |> kbl(digits = 2) |> kable_styling()
```

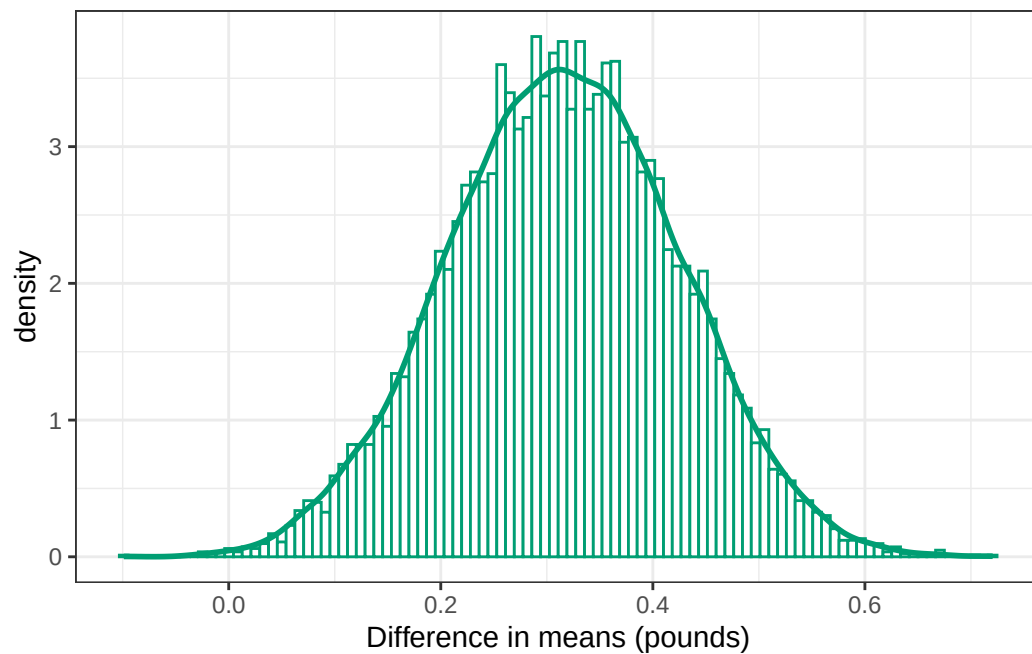
#### 20.1.4 Posterior distribution of difference in means

```
posterior = posterior |>
  mutate(mu_diff = mu1 - mu2)

posterior |> head() |> kbl(digits = 2) |> kable_styling()
```

```
ggplot(posterior, aes(x = mu_diff)) +
  geom_histogram(
    aes(y = after_stat(density)),
    col = bayes_col["posterior"],
    fill = "white",
    bins = 100
  ) +
  geom_density(linewidth = 1, col = bayes_col["posterior"]) +
```

```
labs(x = "Difference in means (pounds)") +  
theme_bw()
```



```
mean(posterior$mu_diff)
```

```
[1] 0.315822
```

```
sd(posterior$mu_diff)
```

```
[1] 0.1082119
```

```
quantile(posterior$mu_diff, c(0.01, 0.10, 0.25, 0.5, 0.75, 0.90, 0.99))
```

|  | 1%         | 10%        | 25%        | 50%        | 75%        | 90%        | 99%        |
|--|------------|------------|------------|------------|------------|------------|------------|
|  | 0.06890393 | 0.17628197 | 0.24189140 | 0.31563945 | 0.39102622 | 0.45509328 | 0.56207534 |

```
sum(posterior$mu_diff > 0) / length(posterior$mu_diff)
```

```
[1] 0.9989
```

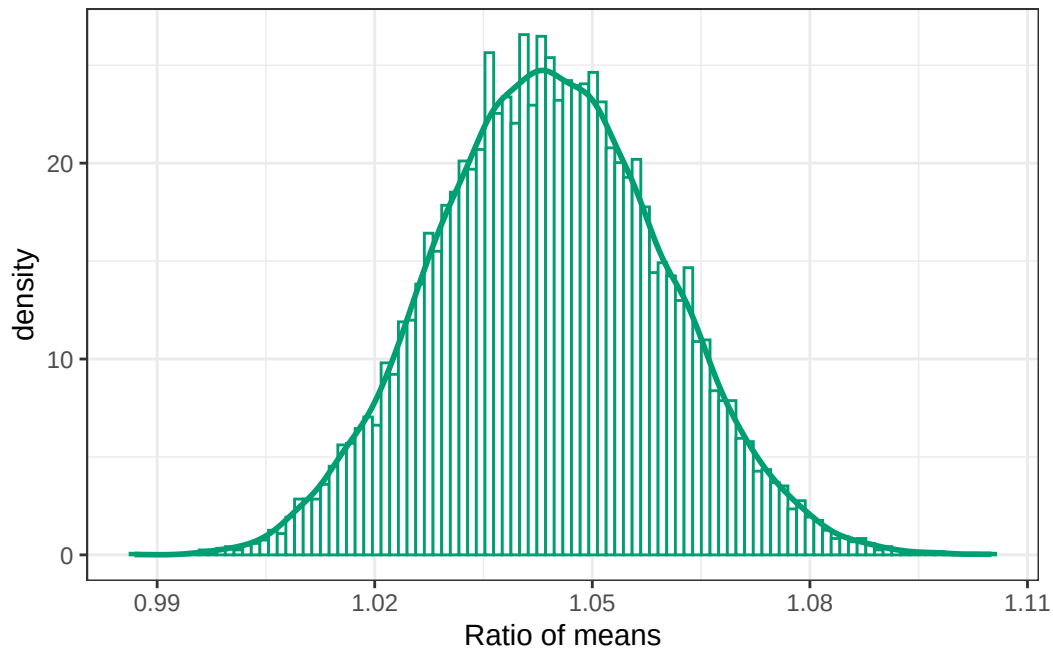
| .chain | .iteration | .draw | mu2  | mu1  | sigma | mu_diff | mu_ratio |
|--------|------------|-------|------|------|-------|---------|----------|
| 1      | 1          | 1     | 7.34 | 7.49 | 1.09  | 0.15    | 1.02     |
| 1      | 2          | 2     | 7.27 | 7.50 | 1.04  | 0.23    | 1.03     |
| 1      | 3          | 3     | 7.24 | 7.51 | 1.06  | 0.27    | 1.04     |
| 1      | 4          | 4     | 7.23 | 7.50 | 1.05  | 0.27    | 1.04     |
| 1      | 5          | 5     | 7.32 | 7.44 | 1.09  | 0.13    | 1.02     |
| 1      | 6          | 6     | 7.08 | 7.52 | 1.09  | 0.45    | 1.06     |

### 20.1.5 Posterior distribution of ratio of means

```
posterior = posterior |>
  mutate(mu_ratio = mu1 / mu2)

posterior |> head() |> kbl(digits = 2) |> kable_styling()
```

```
ggplot(posterior, aes(x = mu_ratio)) +
  geom_histogram(
    aes(y = after_stat(density)),
    col = bayes_col["posterior"],
    fill = "white",
    bins = 100
  ) +
  geom_density(linewidth = 1, col = bayes_col["posterior"]) +
  labs(x = "Ratio of means") +
  theme_bw()
```



```
mean(posterior$mu_ratio)
```

```
[1] 1.044161
```

```
sd(posterior$mu_ratio)
```

```
[1] 0.01564791
```

```
quantile(posterior$mu_ratio, c(0.01, 0.10, 0.25, 0.5, 0.75, 0.90, 0.99))
```

|  | 1%       | 10%      | 25%      | 50%      | 75%      | 90%      | 99%      |
|--|----------|----------|----------|----------|----------|----------|----------|
|  | 1.009344 | 1.024155 | 1.033327 | 1.043923 | 1.054902 | 1.064448 | 1.080669 |

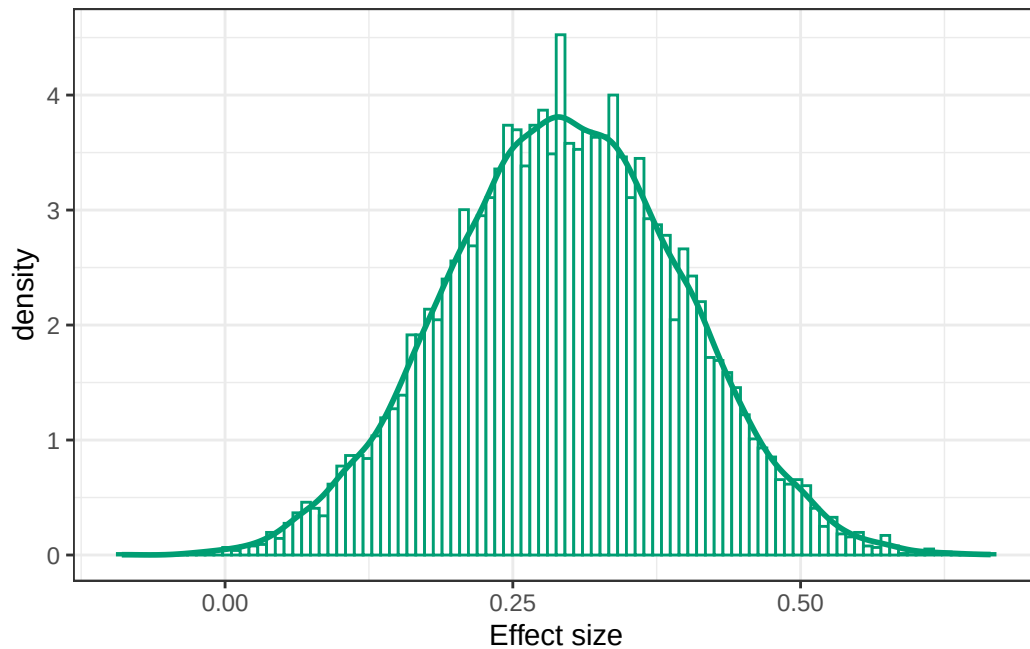
## 20.1.6 Posterior distribution of effect size

```
posterior = posterior |>
  mutate(effect_size = (mu1 - mu2) / sigma)

posterior |> head() |> kbl(digits = 2) |> kable_styling()
```

| .chain | .iteration | .draw | mu2  | mu1  | sigma | mu_diff | mu_ratio | effect_size |
|--------|------------|-------|------|------|-------|---------|----------|-------------|
| 1      | 1          | 1     | 7.34 | 7.49 | 1.09  | 0.15    | 1.02     | 0.13        |
| 1      | 2          | 2     | 7.27 | 7.50 | 1.04  | 0.23    | 1.03     | 0.22        |
| 1      | 3          | 3     | 7.24 | 7.51 | 1.06  | 0.27    | 1.04     | 0.25        |
| 1      | 4          | 4     | 7.23 | 7.50 | 1.05  | 0.27    | 1.04     | 0.26        |
| 1      | 5          | 5     | 7.32 | 7.44 | 1.09  | 0.13    | 1.02     | 0.11        |
| 1      | 6          | 6     | 7.08 | 7.52 | 1.09  | 0.45    | 1.06     | 0.41        |

```
ggplot(posterior, aes(x = effect_size)) +
  geom_histogram(
    aes(y = after_stat(density)),
    col = bayes_col["posterior"],
    fill = "white",
    bins = 100
  ) +
  geom_density(linewidth = 1, col = bayes_col["posterior"]) +
  labs(x = "Effect size") +
  theme_bw()
```



```
mean(posterior$effect_size)
```

```
[1] 0.2952176
```



```
sd(posterior$effect_size)
```

```
[1] 0.1015268
```

```
quantile(posterior$effect_size, c(0.01, 0.10, 0.25, 0.5, 0.75, 0.90, 0.99))
```

|  | 1%         | 10%        | 25%        | 50%        | 75%        | 90%        | 99%        |
|--|------------|------------|------------|------------|------------|------------|------------|
|  | 0.06374755 | 0.16458943 | 0.22577629 | 0.29451560 | 0.36465077 | 0.42560646 | 0.52678586 |